

SALLA VARRIOTUNTURI, Finland

WMO: 028190

Lat: 67.749N

Lon: 29.611E

Elev: 1181

StdP: 14.08

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.3	(c) -5.9	(d) -15.2	(e) 2.5	(f) -10.1	(g) -10.4	(h) 3.3	(i) -5.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	75.4	59.9	71.2	58.6	67.6	57.2	62.8	70.9	60.5	69.0	58.3	65.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	79.1	65.8	57.0	72.4	62.9	55.1	67.4	60.8	29.0	71.0	27.3	69.2	25.8	65.1	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -16.0	(f) 77.9	(g) 4.8	(h) 4.7	(i) -19.4	(j) 81.3	(k) -22.3	(l) 84.0	(m) -25.0	(n) 86.6	(o) -28.5	(p) 90.0		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-15.6	65.8	5.2	3.5	-19.3	68.3	-22.3	70.3	-25.2	72.2	-29.0	74.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.5	10.8	13.2	19.9	29.8	40.9	49.2	56.5	52.3	44.3	31.3	22.3	17.7		
	DBStd	17.22	9.37	10.32	7.59	6.86	8.62	8.53	8.00	7.11	6.35	6.58	7.62	8.66		
	HDD50	6895	1214	1030	934	607	306	117	29	57	190	579	831	1001		
	HDD65	11908	1679	1450	1399	1057	746	481	283	400	622	1044	1281	1466		
	CDD50	492	0	0	0	0	26	92	229	127	18	0	0	0		
	CDD65	30	0	0	0	0	1	6	18	5	0	0	0	0		
	CDH74	171	0	0	0	0	4	34	104	29	0	0	0	0		
	CDH80	19	0	0	0	0	0	1	13	5	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	23.5	1.5	1.3	1.2	1.4	1.9	2.6	3.1	2.6	2.3	2.2	1.6	1.6		
(15) Precipitation	PrecMax	29.6	2.8	3.2	2.4	3.0	3.0	4.6	5.1	5.0	4.1	4.6	3.1	2.8		
	PrecMin	17.7	0.5	0.5	0.3	0.5	1.0	0.6	1.4	0.6	0.6	0.7	0.4	0.4		
	PrecStd	3.2	0.6	0.7	0.6	0.6	0.6	1.2	1.1	1.2	1.0	1.0	0.7	0.6		
	PrecStd	3.2	0.6	0.7	0.6	0.6	0.6	1.2	1.1	1.2	1.0	1.0	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	34.7	39.3	54.5	72.8	78.5	82.0	77.8	62.5	48.0	38.7	34.1		
		MCWB	31.0	33.7	34.4	44.3	53.8	60.5	63.2	63.8	54.0	47.4	37.9	33.0		
	2%	DB	28.4	31.8	35.1	45.6	65.9	72.5	76.7	71.1	58.8	44.1	35.9	32.1		
		MCWB	27.8	31.0	32.1	38.3	51.9	59.0	61.0	60.1	52.4	42.3	35.3	31.2		
	5%	DB	25.5	29.7	32.3	42.1	60.3	66.9	72.8	67.2	56.0	42.2	33.5	30.7		
		MCWB	24.8	29.0	30.2	36.1	48.8	54.9	58.9	58.1	51.1	40.9	33.0	30.1		
	10%	DB	23.2	27.2	30.5	39.3	54.7	62.7	69.3	63.2	53.5	39.8	32.2	28.8		
		MCWB	22.5	26.5	28.9	34.4	45.7	52.2	58.4	55.7	49.4	38.9	31.8	28.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	34.0	34.9	43.3	55.4	63.0	66.6	65.7	56.4	47.4	38.4	33.4		
		MCDB	31.2	34.5	37.9	53.4	67.3	74.2	73.7	76.5	59.6	47.7	38.6	33.9		
	2%	WB	27.9	31.3	32.6	39.2	52.6	60.2	63.7	61.5	53.5	42.8	35.1	31.5		
		MCDB	28.4	31.7	34.5	44.3	64.7	70.2	71.3	68.7	57.0	43.5	35.6	32.0		
	5%	WB	25.0	29.1	30.7	36.7	49.6	56.5	61.4	58.5	51.9	40.8	33.2	30.1		
		MCDB	25.4	29.7	32.0	40.9	59.1	63.7	69.7	65.2	55.4	42.0	33.5	30.7		
	10%	WB	22.7	26.6	29.0	35.2	46.4	53.9	59.6	56.8	50.0	39.1	31.8	28.2		
		MCDB	23.2	27.2	30.3	38.4	53.5	61.0	67.4	61.8	53.0	39.8	32.1	28.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	8.6	10.3	11.7	13.4	13.6	13.6	12.0	9.3	6.2	7.1	7.8		
		MCDBR	10.2	8.4	9.4	13.9	21.8	18.2	18.2	16.5	12.5	7.3	7.0	6.9		
		MCWBR	10.2	8.4	7.8	8.9	11.7	9.3	8.2	8.5	7.9	6.6	6.9	6.9		
	5% WB	MCDBR	10.0	8.3	8.4	12.3	20.6	16.2	15.9	14.6	10.9	6.9	7.0	6.7		
		MCWBR	10.1	8.4	7.4	7.9	11.7	9.5	8.4	8.5	7.9	6.4	7.0	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.190	0.248	0.288	0.317	0.337	0.338	0.352	0.333	0.304	0.268	0.354	N/A		
	taud		1.913	2.080	2.147	2.192	2.387	2.467	2.448	2.475	2.534	2.299	2.748	N/A		
	Ebn at Noon		38	176	236	263	272	275	266	259	240	173	53	0		
	Edh at Noon		3	16	26	33	31	30	29	25	19	12	4	0		
(44) All-Sky Solar Radiation	RadAvg		14	162	619	1156	1485	1571	1432	1024	547	174	32	0		
	RadStd		1	21	47	74	149	131	160	99	49	29	4	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page